MATH 261, Homework Set 7 Name: _____

1. Let $\mathbf{F} = 2xy\,\mathbf{i} + (x^2 + y^2)\,\mathbf{j}$.
   (a) Compute the divergence of $\mathbf{F}$.
   (b) Compute the k-component of the curl of $\mathbf{F}$.
   (c) Is this vector field conservative?

(a) Divergence	**(b) k-component of curl**	**(c) Conservative?**

2. Let $\mathbf{F} = \sin(x^2)\,\mathbf{i} + (x + e^{y^2})\,\mathbf{j}$.
   (a) Compute the divergence of $\mathbf{F}$.
   (b) Compute the k-component of the curl of $\mathbf{F}$.
   (c) Is this vector field conservative?

(a) Divergence	**(b) k-component of curl**	**(c) Conservative?**

3. Let $\mathbf{F} = x^2\,\mathbf{i} + z^2\,\mathbf{j} + xy\,\mathbf{k}$.

   (a) Compute the divergence of $\mathbf{F}$.

   (b) Compute the curl of $\mathbf{F}$.

   (c) Is this vector field conservative?

(a) Divergence	**(b) Curl**	**(c) Conservative?**

4. Let $\mathbf{F} = x^3\,\mathbf{k}$.

   (a) Compute the divergence of $\mathbf{F}$.

   (b) Compute the curl of $\mathbf{F}$.

   (c) Is this vector field conservative?

(a) Divergence	**(b) Curl**	**(c) Conservative?**

5. Let $\mathbf{F} = yz\,\mathbf{i} + xz\,\mathbf{j} + xy\,\mathbf{k}$.

   (a) Compute the divergence of $\mathbf{F}$.

   (b) Compute the curl of $\mathbf{F}$.

   (c) Is this vector field conservative?

(a) Divergence	**(b) Curl**	**(c) Conservative?**

6. Let $\mathbf{F}(x, y, z) = z\cos(y)\,\mathbf{i} - xz\sin(y)\,\mathbf{j} + x\cos(y)\,\mathbf{k}$.

   (a) Compute the divergence of $\mathbf{F}$.

   (b) Compute the curl of $\mathbf{F}$.

   (c) Is this vector field conservative?

(a) Divergence	**(b) Curl**	**(c) Conservative?**

7. Find the curl and divergence for $\mathbf{F}(x, y, z) = x^2 z\,\mathbf{i} - 2y^2 z^2\,\mathbf{j} + xy^2 z\,\mathbf{k}$ at the point $(1, -1, 1)$.

Curl	**Divergence**

8. Find the curl and divergence for $\mathbf{F}(x, y, z) = xz^3\,\mathbf{i} - 2x^2 yz\,\mathbf{j} + 2yz^4\,\mathbf{k}$ at $(1, -1, 1)$.

Curl	**Divergence**

9. Find a potential function for $\mathbf{F} = 2x\,\mathbf{i} + 3\,\mathbf{j} + 4z\,\mathbf{k}$.

Potential function

10. Find a potential function for $\mathbf{F} = y^2\,\mathbf{i} + (2xy + e^z)\,\mathbf{j} + ye^z\,\mathbf{k}$.

Potential function

11. Find a potential function for $\mathbf{F} = \cos z\,\mathbf{i} + 2y\,\mathbf{j} - x\sin(z)\,\mathbf{k}$.

Potential function

12. Use FTOLI to evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

where $\mathbf{F} = 2xyz\,\mathbf{i} + x^2z\,\mathbf{j} + x^2y\,\mathbf{k}$ and $\mathbf{r}(t) = \langle t^2,\ \sin(\pi t/4),\ e^{t^2-2t}\rangle$, $0 \le t \le 2$.

Line integral

13. Use FTOLI to evaluate

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

where $\mathbf{F} = x\,\mathbf{i} + z\cos(y)\,\mathbf{j} + \sin(y)\,\mathbf{k}$ and C is the ellipse $4x^2 + 9y^2 = 36$, traversed clockwise.

Line integral

14. Let $\mathbf{F} = (x-y)\mathbf{i} + (x+y)\mathbf{j}$ and let C be the circle with radius 1 traversed counterclockwise. Verify Green's Theorem for $\int_C \mathbf{F} \cdot \mathbf{T} ds$ (First compute the line integral directly then use Green's Theorem and verify that both methods give you the same answer)

Direct line integral

Green's Theorem result

15. Let $\mathbf{F} = y\mathbf{i} + x\mathbf{j}$ and let C be the triangle with vertices $(0,0), (1,0), (1,1)$ traversed counterclockwise. Verify Green's Theorem for $\int_C \mathbf{F} \cdot \mathbf{T} ds$.

Direct line integral	**Green's Theorem result**

16. Use Green's Theorem to compute $\int_C \mathbf{F} \cdot \mathbf{T} ds$, $\mathbf{F} = xy^5\,\mathbf{i} + (y^2 + x^2)\mathbf{j}$ on the rectangle with vertices $(0,0), (0,1), (1,0), (1,1)$, traversed counterclockwise.

Circulation

17. Now compute $\int_C \mathbf{F} \cdot \mathbf{N} ds$ for the same $\mathbf{F}$ and rectangle of the previous exercise.

Flux

18. Use Green's Theorem to compute $\int_C \mathbf{F} \cdot \mathbf{T} ds$, $\mathbf{F} = ye^x \, \mathbf{i} + (y^2 + x^2)\mathbf{j}$ on the rectangle with vertices $(0, 0), (0, 3), (5, 0), (5, 3)$, traversed counterclockwise.

Circulation

19. Repeat the previous exercise but now compute $\int_C \mathbf{F} \cdot \mathbf{N} ds$ instead.

Flux

20. Use Green's Theorem to evaluate $\int_C \mathbf{F} \cdot \mathbf{T} ds$, $\mathbf{F} = (x^2 + y)\mathbf{i} + (y^2 - x)\mathbf{j}$ on the triangle $(0, 0), (-1, 0), (-1, 1)$, traversed counterclockwise.

Circulation

21. Repeat the previous exercise but now compute $\int_C \mathbf{F} \cdot \mathbf{N} ds$ instead.

Flux

22. Use Green's Theorem to find the work done by $\mathbf{F}(x, y) = (x + y)\mathbf{i} + xy^2\mathbf{j}$ in moving a particle from the origin along the x-axis to (1,0), then along a line segment to (0,1), and then back to the origin along the y-axis.

Work

23. Use Green's Theorem to find the work done by $\mathbf{F}(x, y) = x^2y\,\mathbf{i} + xy^2\mathbf{j}$ in moving a particle down from the origin along the line $y = -x$ to the point (1,-1), then up along a line segment to (1,1), and then back to the origin along the line $y = x$.

Work

24. Use Green's Theorem to find the flux of the vector field $\mathbf{F}(x, y) = x^2\mathbf{i} + y^2\mathbf{j}$ across the circle $x^2 + y^2 = 1$.

Flux

25. Use Green's Theorem to find the flux of $\mathbf{F}(x, y) = x\,\mathbf{i} + y^2\,\mathbf{j}$ across the curve obtained by following the graph of $y = x^2$ from (0,0) to (1,1), the line $y = 1$ from (1,1) to (0,1), and the line $x = 0$ from (0,1) to (0,0).

Flux